Data Visualisation Project Design Book

COS30045 - Data Visualisation

Tutor: Andrew Guthrie

Student:	***ID:***
Hoang Trang Anh Pham	105006870
Phuong Bao Minh Nguyen	104339685

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Contents

1 Introduction **2**

1.1 Background and Motivation . . . 2

1.2 Visualisation Purpose . . . 3

2 Data **5**

2.1 Data Source and Governance . . . 5

2.2 Data Processing and Analysis . . . 6

2.3 Data Exploration . . . 8

3 Visualisation Design **10**

3.1 Website Design . . . 10

3.2 Visualisation Design . . . 13

3.2.1 Chart type choices . . . 13

3.2.2 Design Principles . . . 17

3.3 Interaction Design . . . 18

4 Iteration and Validation **23**

4.1 Testing and refinements . . . 23

4.2 Usability evaluation . . . 26

5 Conclusion and Future Improvements **27**

5.1 Conclusion . . . 27

5.2 Future Improvements . . . 28

6 Appendix 1: GenAI Usage **29**

7 Appendix 2: Resources Used **29**

8 Reference **29**

1 Introduction

1.1 Background and Motivation

This project addresses the need for accessible, detailed, and actionable insights into Australian road safety enforcement data. The Bureau of Infrastructure and Transport Research Economics (BITRE) has been a central authority in publishing Australian transport statistics since 1970 and began collecting road safety enforcement data in 2008. In 2023, BITRE introduced more granular information, capturing attributes such as location, age group, arrests, and charges.

While BITRE's existing Police Enforcement Dashboard presents summary statistics using Power BI, it does not incorporate this new level of detail. This limits users' ability to explore emerging patterns and answer more targeted questions about road safety enforcement. The current project was initiated to overcome these limitations by designing and implementing an interactive web-based dashboard, enabling users to engage with the most recent, fine-grained dataset in a more flexible and insightful manner.

The initial goal for this project was to leverage all available data, including the newly added location information (remoteness). However, analysis revealed that not all jurisdictions collect or report location and remoteness in a consistent manner. Manual data entry practices resulted in significant gaps and inconsistencies, making meaningful spatial analysis unfeasible. Based on these limitations, the project scope was refined to focus on consistently reported aspects of the data. The visualisation therefore excludes the location variable and instead concentrates on questions supported by robust data from all jurisdictions.

The key questions addressed by this visualisation are:

- What are the overall statistics for speeding infringements, fines, arrests, and charges?
- How do speeding fines vary across detection methods throughout 2023?
- Which jurisdictions and age groups contribute the most to the total fines?
- Which jurisdictions have the highest rates of speeding fines when adjusted for driver licenses?
- How do enforcement activities and demographic patterns change over time?

The target audience for this visualisation includes policymakers, law enforcement agencies, transport analysts, and members of the public who seek greater transparency and understanding of road safety enforcement practices across Australia.

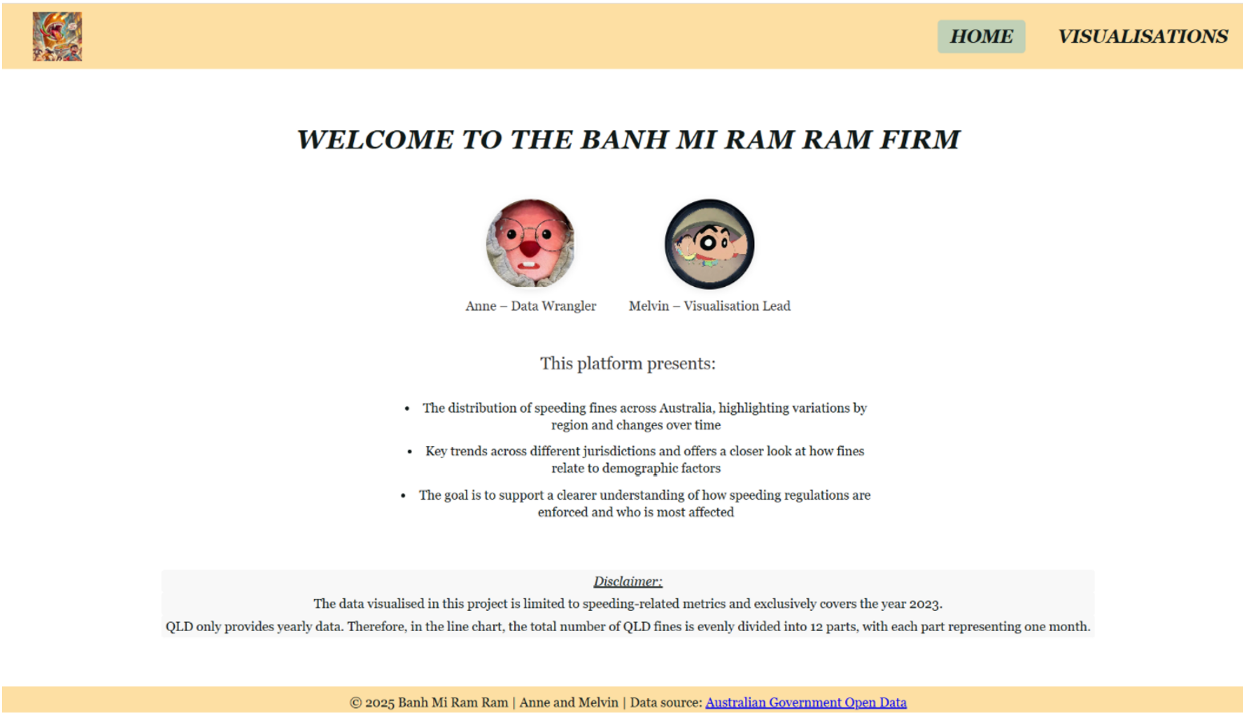

Figure 1. Notes and Disclaimers on the Dataset

1.2 Visualisation Purpose

The purpose of this project is to create an interactive dashboard that enables users to explore, compare, and interpret Australian road safety enforcement data from multiple perspectives. The dashboard provides both an overview and the ability to focus on specific dimensions, such as jurisdiction or age group, through a single global jurisdiction filter and a suite of coordinated charts.

By using this dashboard, users can:

- View overall statistics for speeding infringements, fines, arrests, and charges.
- Analyse how speeding fines differ by detection method during 2023.
- Identify which jurisdictions and age groups account for the largest share of fines.
- Determine which regions issue the highest rates of speeding fines when adjusted for the number of driver licenses.
- Observe how enforcement activities and demographic patterns change throughout the year.

This tool is intended to make complex enforcement data accessible and actionable for a wide audience. It supports decision-making, transparency, and public engagement by providing clear, meaningful visual narratives and by encouraging users to explore the underlying data in detail.

Benefits and Impact of the Visualisation:

- The global jurisdiction filter ensures a unified and consistent experience, with all main charts updating together in response to user selections. This enables efficient, focused exploration of the data.
- Each chart is tailored to communicate a specific message, allowing users to quickly understand trends, distributions, and comparisons relevant to enforcement activity.
- Interactive tooltips and clear visual marks provide immediate access to detailed information, increasing both the depth and accessibility of the analysis.
- The dashboard helps analysts and policymakers streamline their comparison and decision-making process, while also presenting the data in an engaging format for the public.
- The design uses a visually accessible, colour-blind friendly palette and is fully responsive for use across a variety of devices, making it inclusive for users with different needs.

By focusing on reliable data attributes and prioritising user-friendly design, this project offers a practical and insightful resource for understanding and improving road safety enforcement across Australia.

2 Data

2.1 Data Source and Governance

The original data source is the Road Safety Enforcement Data published by the Bureau of Infrastructure and Transport Research Economics (BITRE). This dataset is publicly accessible and made available through the official BITRE website. The data provides insights into traffic enforcement activity across Australia, including fines and infringement notices issued by police and transport authorities in various states and territories. The 2024 edition of the dataset can be accessed at this link.

Data summary table:

Attribute	**Description**
Data source	Bureau of Infrastructure and Transport Research Economics (BITRE)
Number of records	5,017
Jurisdictions	8 (ACT, NSW, NT, QLD, SA, TAS, VIC, WA)
Key attributes	YEAR, START_DATE, JURISDICTION, LOCATION, AGE_GROUP, METRIC, DETECTION_METHOD, FINES, ARRESTS, CHARGES
Data period	January 2023 – December 2023
Update frequency	Annual (latest: 2023)
Known limitations	QLD lacks monthly and age group breakdowns; some location data missing

In terms of data governance, the data collection process involves BITRE working in collaboration with state and territory transport departments, police forces, and road enforcement agencies. Each jurisdiction submits annual enforcement data in a standardized format, ensuring consistency across the dataset. BITRE then compiles and harmonizes the data, resolving differences in reporting standards and definitions where necessary. To ensure high data quality, BITRE conducts validation checks, such as verifying the alignment between offence codes and descriptions, identifying duplicate entries, and checking for missing or anomalous values. This quality assurance processes help maintain the dataset's reliability and usability for analysis.

Security, privacy, and ethical considerations are integral to the management of this dataset. The data is stored and served through secure Australian Government platforms, ensuring protection against unauthorized access or tampering. Importantly, the dataset is fully de-identified and does not include any personal information such as names, addresses, or vehicle registration numbers. The data is aggregated in a way that prevents the re-identification of individuals or specific events. From an ethical standpoint, the dataset is intended to inform public policy, academic research, and public awareness, rather than to assign blame or stigmatize particular regions or communities. Users of the data are expected to adhere to principles of responsible use and are encouraged to accurately cite BITRE as the source.

2.2 Data Processing and Analysis

Key attributes and their data types:

Key attributes	**Original type**	**Data type**
YEAR	Number (integer)	Interval
START_DATE	String	Interval (Date)
END_DATE	String	Interval (Date)
JURISDICTION	String	Nominal (Categorical)
LOCATION	String	Nominal (Categorical)
AGE_GROUP	String	Nominal (Categorical)
METRIC	String	Nominal (Categorical)
DETECTION_METHOD	String	Nominal (Categorical)
FINES	Number (double)	Ratio (Quantitative)
ARRESTS	Number (double)	Ratio (Quantitative)
CHARGES	Number (double)	Ratio (Quantitative)

The scope of this project is limited to analysing speeding-related metrics for the year 2023. Consequently, all records pertaining to other offence types and years were excluded during the data preprocessing stage. The 'Location' attribute was also omitted from analysis, as it is only available in 3 out of the 8 jurisdictions (ACT, NSW, and VIC). Moreover, the data from the ACT lacks meaningful spatial granularity, with all entries uniformly labelled as "Major Cities of Australia," rendering it uninformative for regional analysis. To ensure consistency and relevance, the dataset was subsequently segmented and processed separately for each jurisdiction using KNIME.

A notable challenge encountered was the inconsistency in data completeness across jurisdictions. For instance, in Victoria during January 2023, no offences were recorded for individuals aged 0–16, resulting in the absence of rows for this age group in that period. Additionally, although seven speeding detection methods are recognised nationally, individual jurisdictions such as Victoria reported only three. These disparities hinder the use of the dataset as a reliable basis for cross-jurisdictional comparison.

To address these inconsistencies, a standardisation approach was implemented. Following group discussions, it was agreed that representing unreported combinations with zero values (rather than omitting them) would better support interpretability and comparative analysis. Using KNIME's GroupBy and Cross Joiner nodes, a complete structure was generated wherein each jurisdiction contains data for every combination of 12 months, 6 age groups, and 7 detection methods—resulting in 504 records per jurisdiction. Newly generated records lacking original data were populated with zero values in the 'Fines', 'Arrests', and 'Charges' fields to indicate the absence of reported offences. Finally, duplicate entries were identified and removed using the Duplicate Row Filter node, with the first occurrence retained in cases of duplication.

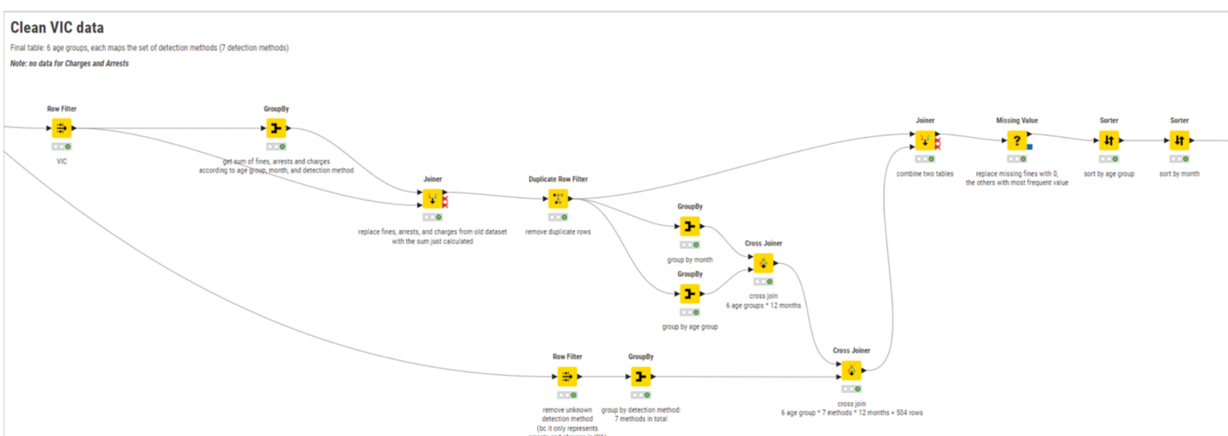

Figure 2. VIC Data Cleaning Workflow

The same method is applied to every jurisdiction.

2.3 Data Exploration

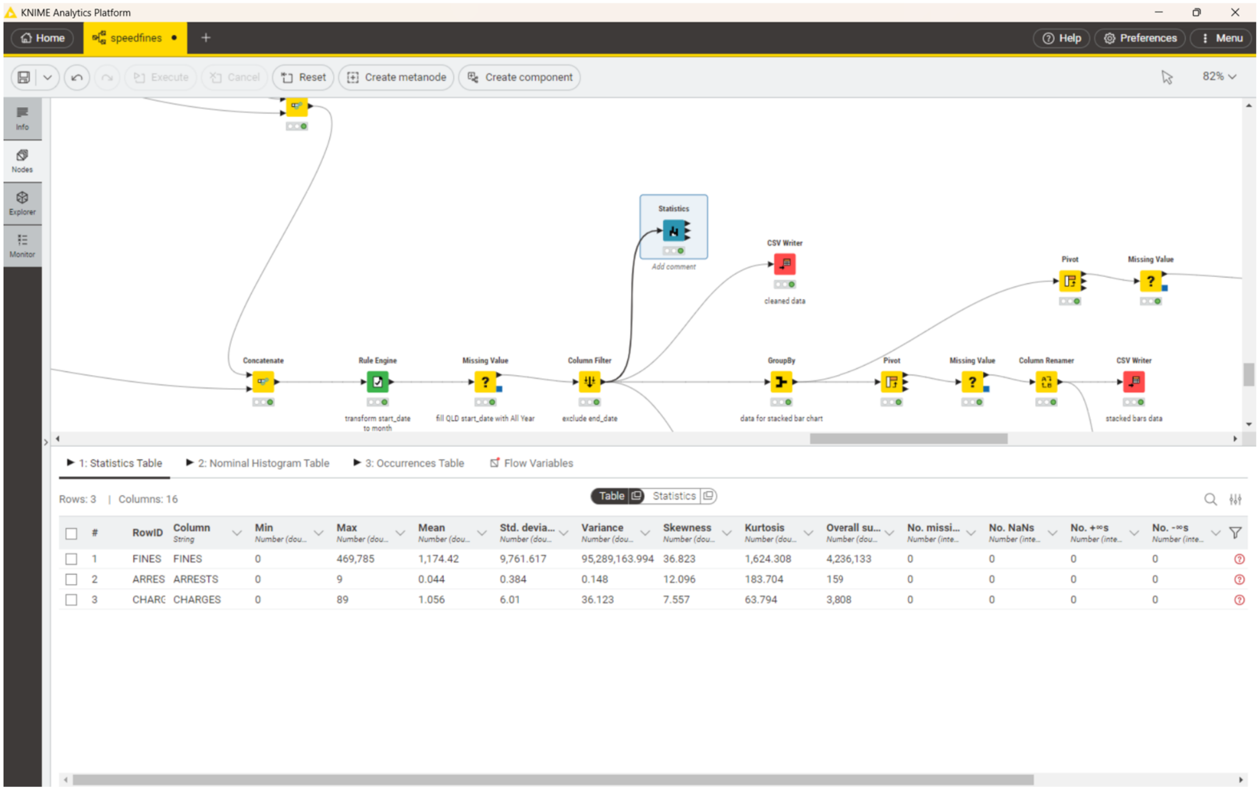

Figure 3. Statistics Node

We used the Statistics node to gain statistical insights from our cleaned dataset. This statistical summary provides insight into the distribution and variability of three enforcement metrics: fines, arrests, and charges.

Among these, fines clearly stand out as the most common enforcement, with over 4.2 million cases recorded. The average (mean) number of fines per record is approximately 1,174. However, the data is highly skewed (skewness = 36.82, standard deviation = 9,761.62), indicating that the distribution is heavily concentrated at the lower end with a small number of records exhibiting extremely high values—up to a maximum of 469,785 fines.

Arrests, on the other hand, are extremely rare — only 159 in total across all jurisdictions and age groups. The average number of arrests per record is very low (0.044), and most entries report zero arrests. The data is highly skewed (12.10), which tells us that arrests, when they do occur, are concentrated in just a few records (with up to 9 arrests in one case). Such low record is inevitable, especially since arrests data isn't applicable for jurisdictions like New South Wales and Tasmania, as noted in the BITRE data dictionary.

Charges show a similar pattern to arrests but occur slightly more often, with a total of 3,808 cases. The mean is 1.056 charges per record, and the maximum recorded is 89 charges. The data remains positively skewed (skewness = 7.56), indicating the presence of outliers. The standard deviation of 6.01 suggests that while most records have low counts, a few have significantly higher numbers of charges.

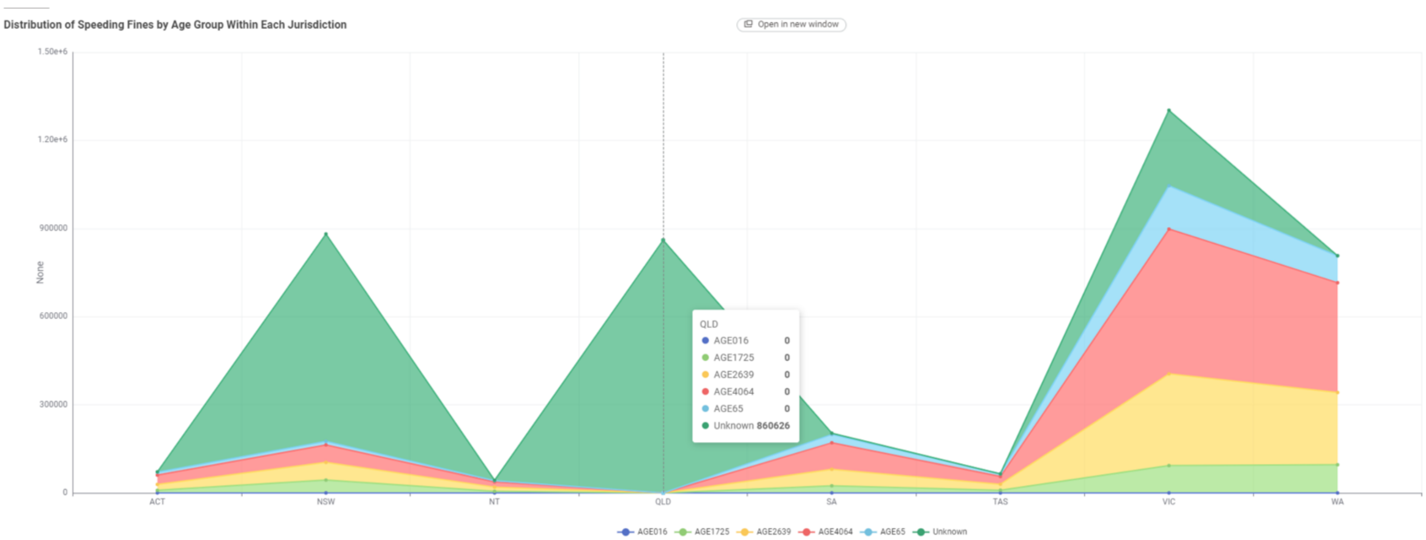

Figure 4. KNIME Stacked Area Chart

From the graph above, it's clear that age group data is not available for Queensland, as all entries from this jurisdiction are marked as "Unknown." Additionally, aside from the four most populated jurisdictions – New South Wales (NSW), Queensland (QLD), Victoria (VIC), and Western Australia (WA) – the number of speeding offences recorded in 2023 is relatively low in other states and territories, with none exceeding 200,000 cases. This disparity is likely influenced by population differences across jurisdictions. To account for this imbalance and provide a more objective view, we chose to create a normalised bar chart to better represent the distribution of fines.

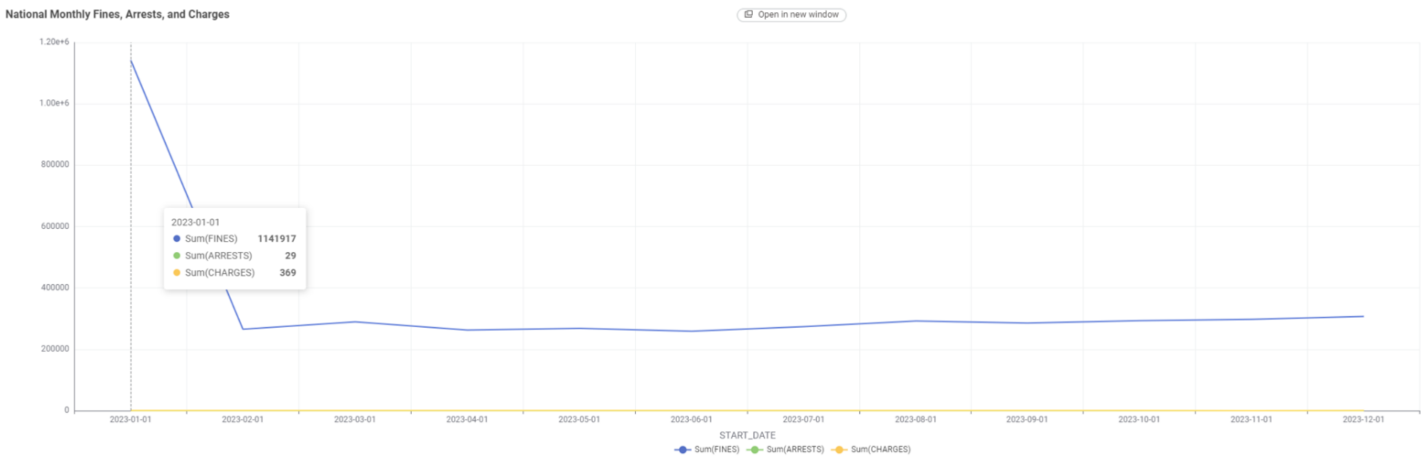

Figure 5. KNIME Line Chart

Not only is Queensland the only jurisdiction that does not categorise enforcement data by age group, this jurisdiction also does not record data on a monthly basis. In the line chart above, the horizontal axis represents the start date of each record. As a result, January appears to have the highest number of fines, but this is misleading – it occurs because all of Queensland's records are dated in January, not because more fines were issued during that month. This presented a challenge when we tried to create a line chart using D3. While we could have simply plotted Queensland's data as-is—similar to the line chart above—we quickly realised that doing so would convey misleading information. To address this, we decided to evenly distribute Queensland's data across all 12 months, allowing for a more accurate and meaningful comparison with other jurisdictions that report monthly data.

3 Visualisation Design

3.1 Website Design

The website adopts a clean, modular layout to maximise data density and maintain a high data-to-ink ratio, ensuring that every visual mark on the screen efficiently communicates information. The main navigation is positioned at the top for clarity, offering quick access to the Overview, Trends, Regional Comparison, and About sections. Each page is organised into regions: the main chart area, filters (as a side panel or dropdowns), and contextual text/keys that guide interpretation.

Figure 6. Wireframe of Website Layout

Wireframe Overview:

- Header: Contains navigation links and project title.
- Main Panel: Displays primary charts and summary card ticks.
- Sidebar/Filter Panel: Hosts interactive filters (jurisdiction, detection method, time interval).
- Footer: Contains attributions, keys, and palette legend.

All visual components have been layered to avoid occlusion, so that overlapping graphical marks (bars, lines, map regions) remain distinguishable. Key interactive controls are spatially separated from the charts, minimising accidental user interference and supporting accessibility.

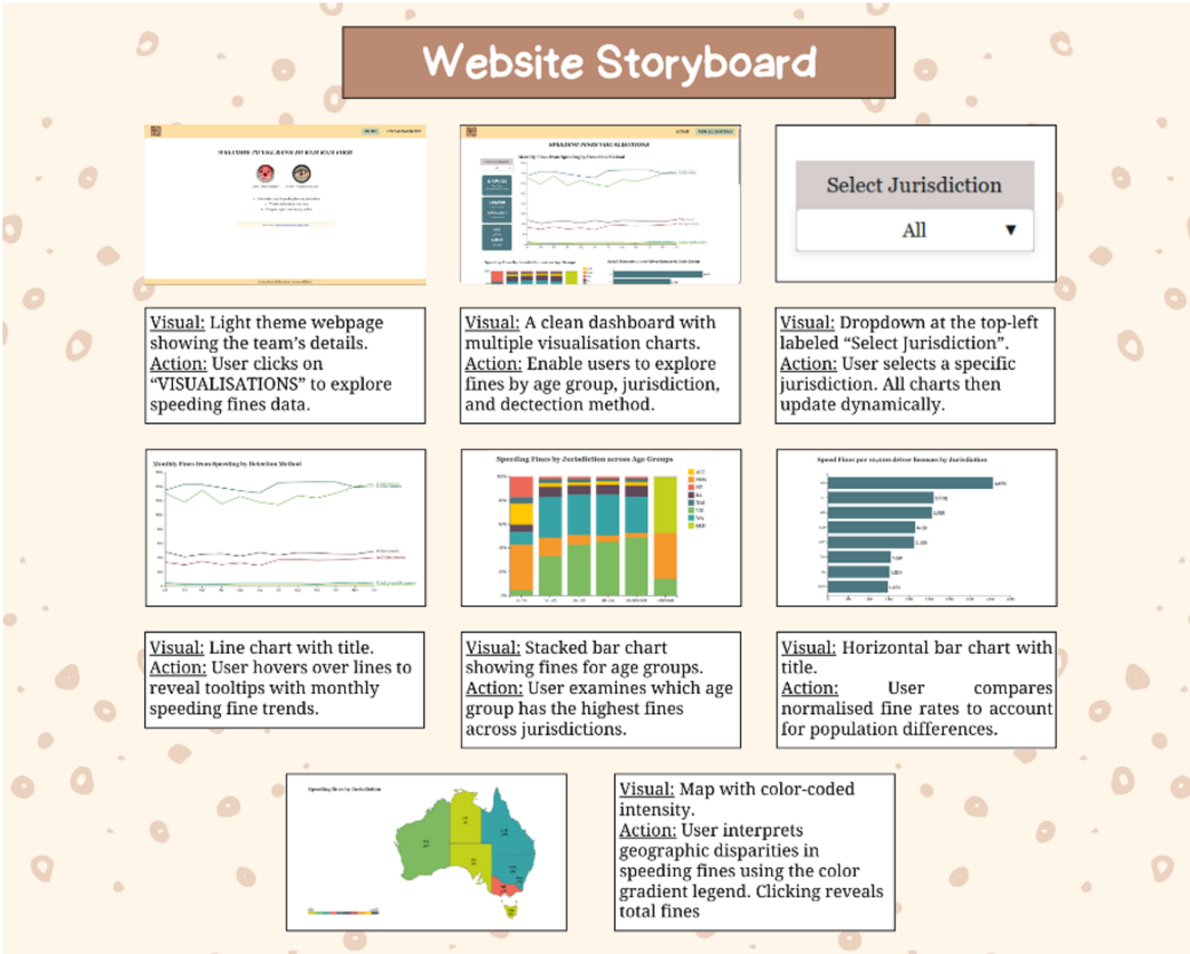

Figure 7. Website Storyboard

Storyboard of User Interaction:

1. Landing: The user visits the visualisation page and is greeted with a summary card section showing total fines, arrests, and charges as visually prominent tick marks.

2. Exploring Trends: Scrolling down, the user sees the line chart displaying monthly speeding fines by detection method. The user hovers over the lines, and detailed tooltips appear, revealing specific values for each month and detection method.

3. Filtering by Jurisdiction: The user opens the filter dropdown labelled "Jurisdiction" in the filter panel. They select one or more jurisdictions. Instantly, all charts—except the map—update to reflect only the selected jurisdictions. Tick marks, hues, and legends dynamically adjust.

4. Comparing Regions: The user scrolls further to the stacked bar and horizontal bar charts. They use the filter to compare different jurisdictions' age group distributions or fine rates,

with updated data-to-ink ratio and clear separation of visual marks.

5. Geographical Exploration: The user clicks on a region in the choropleth map. The map zooms in on the selected jurisdiction, and a detailed tooltip with key region statistics pops up, using saturation and spatial layering to highlight the active region.

6. Resetting Filters: To reset the dashboard to its initial state, the user clicks "Select All" or "Unselect All" in the jurisdiction filter. All charts return to displaying the full dataset, with all ticks, marks, and colours reset.

The design was developed using a visually aesthetic colour palette personally selected to provide sufficient contrast and be suitable for users with colour vision deficiency (tested with online simulators and by using pastel hues and low-saturation tones).

3.2 Visualisation Design

3.2.1 Chart type choices

1. Card Charts (Summary Ticks):

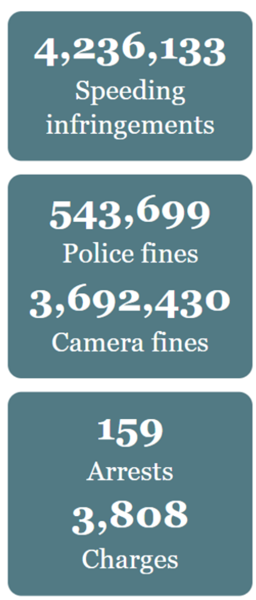

Figure 8. Speeding Fines Value Summary Card Charts

Question: What are the overall stats across speeding infringements for fines, arrests and charts?

Display core headline figures (total fines, arrests, charges) as individual, prominent marks. This maximises cognitive accessibility by providing immediate reference points for the data story. The cards use spatial separation and layering to ensure each metric stands out, without redundant ink or 3D effects that could distort perception.

2. Line Chart

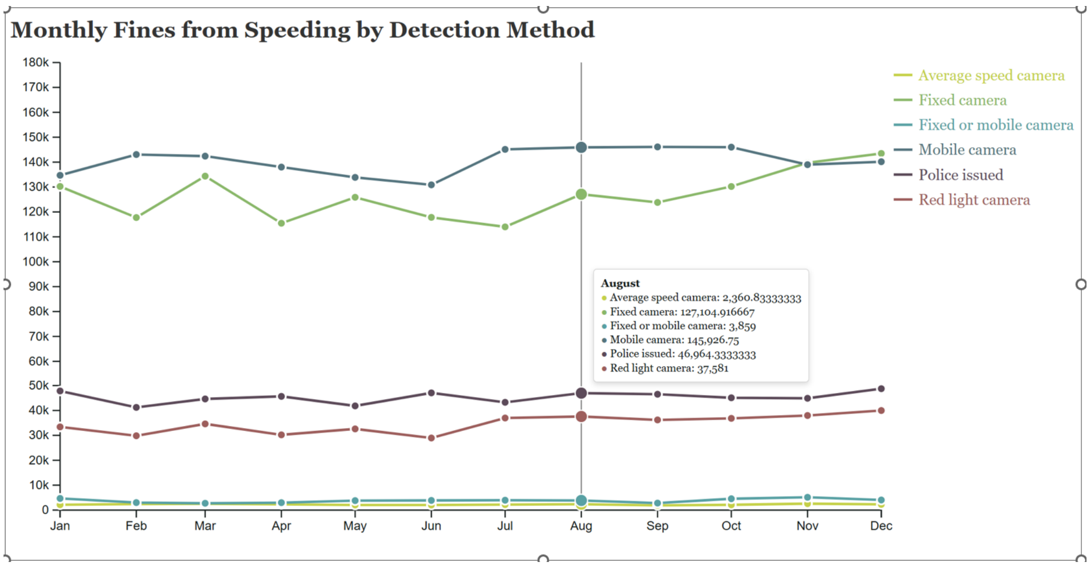

Figure 9. Monthly Fines from Speeding by Detection Method (Line Chart)

Question: How do fines differ across detection methods throughout the months in 2023? Used to visualise monthly fines across different detection methods over time (interval scale). Each line encodes a detection method using a distinct hue, with ticks at each month. This chart type leverages the position and length visual channels for accurate comparison of trends and supports cross-category analysis. The continuous x-axis ensures temporal intervals are faithfully represented, preserving graphical integrity.

3. Stacked Bar Chart

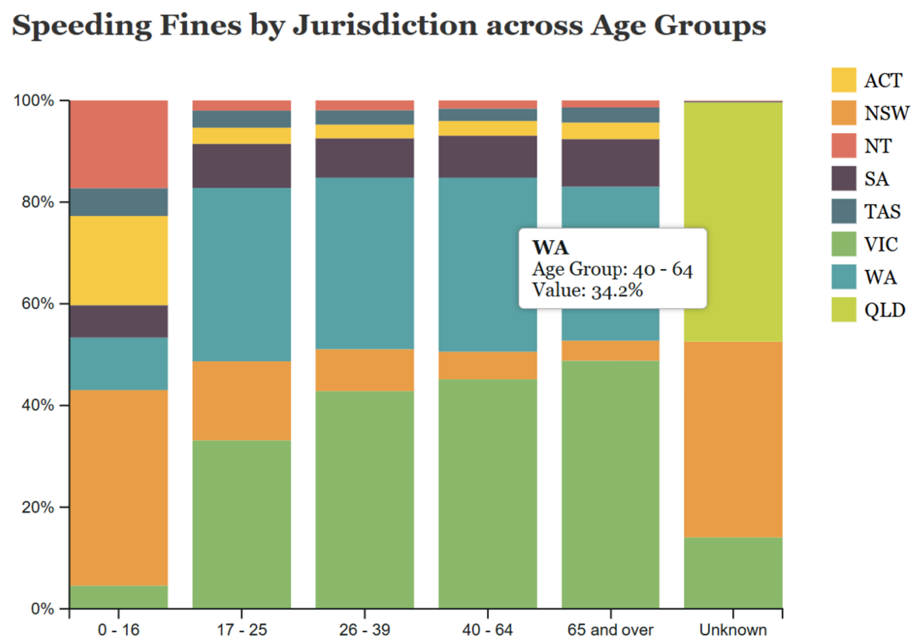

Figure 10. Speeding Fines by Jurisdiction across Age Groups (Stacked Bar Chart)

Question: Which jurisdictions contribute the most speeding fines by age group in Australia in Australia?

Illustrates distribution of fines by age group and jurisdiction. Here, bars are segmented (layered) by jurisdiction (hue) and positioned by age group (ordinal channel). This design enables users to assess both part-to-whole relationships and differences across categories. The chart maintains a high data density while avoiding occlusion by careful arrangement of stacking order and label placement.

4. Horizontal Bar Chart

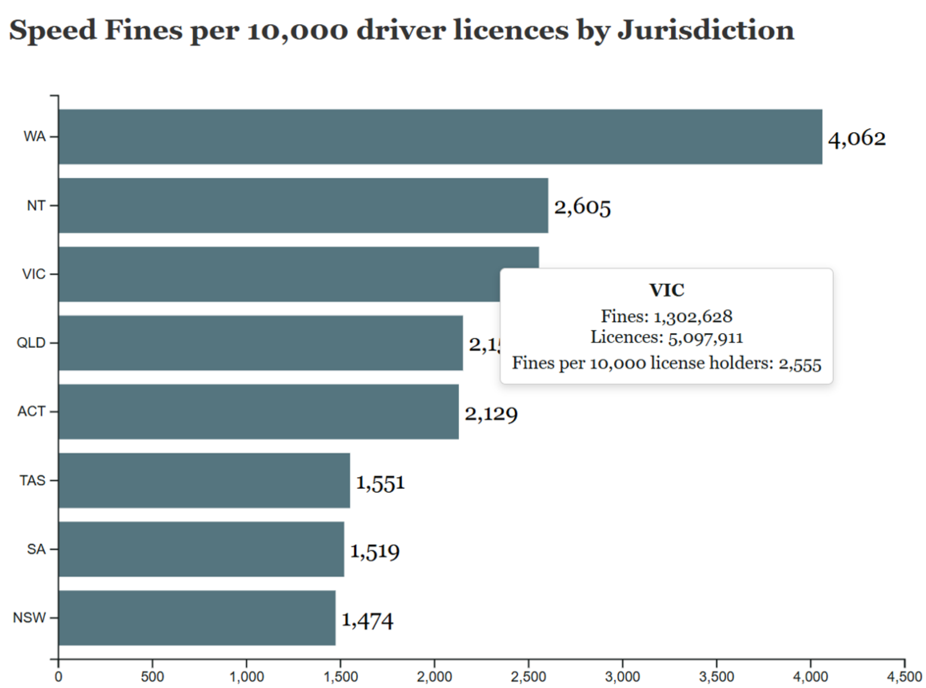

Figure 11. Speed Fines per 10,000 driver licenses by Jurisdiction (Horizontal Bar Chart)

Question: Which jurisdiction has the most speeding fines by driver licenses?

Ranks jurisdictions by fines per 10,000 driver licenses, normalising for population differences and increasing interpretability. Bars are horizontally aligned for easy reading of long jurisdiction names. Marks (bars) are sized by magnitude, with hue indicating jurisdiction, and spatial arrangement supports rapid comparison.

5. Choropleth Map Chart

Figure 12. Speeding Fines Ranking by Jurisdiction (Choropleth Map)

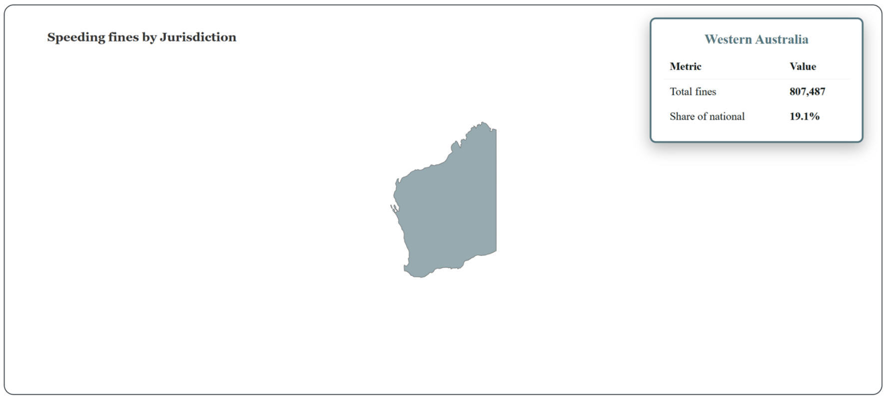

Figure 13. Choropleth Map when clicked in and under tooltip view

Question: How much each jurisdiction's speeding fines accumulate for?

Presents spatial distribution of speeding fines by jurisdiction, with colour saturation representing fine density. Clicking a region activates a detailed tooltip and zoom, using layering to highlight the selected spatial region. The map uses a quantised interval colour scale, tested for accessibility and graphical integrity.

3.2.2 Design Principles

1. Graphical Integrity
   - All axes start at zero where relevant and are clearly labelled.
   - Population-normalised metrics (per 10,000 licenses) are used to avoid misleading users due to raw population differences.
   - Data-to-ink ratio is kept high by removing non-data elements and gridlines that do not aid interpretation.
   - No 3D charts are used, as these often reduce data density and can distort perceived differences.
2. Accessibility
   - Colour palettes are pastel, low-saturation, and carefully chosen to be readable by people with common types of colour blindness (tested using online accessibility tools).
   - Marks are distinguishable by both hue and shape when possible (e.g., different line styles for detection methods).
   - All interactive ticks (buttons, bars, lines) are sized to be usable on both desktop and mobile.

- Font sizes and key region labels meet WCAG minimum standards for contrast and readability.

3. Scalability

   - The modular page layout supports responsive stacking or grid behaviour, ensuring charts remain readable at various screen sizes (media queries, CSS Flexbox/Grid).
   - Legends and keys adapt their spatial arrangement for smaller screens.
   - Data density is preserved regardless of viewport size; tick marks and labels are never allowed to overlap (managed by dynamic label placement and occlusion avoidance).

4. Visual Encoding
   Data attributes are mapped to visual channels in a way that maximises perceptual accuracy:

   - Interval and ratio data are mapped to length and position (line/bar charts).
   - Ordinal categories use spatial separation (x or y position) and consistent order.
   - Region is encoded using both position (map) and hue (jurisdiction key).
   - Magnitude is mapped to bar length or colour saturation, never area or volume.

   Layering and separation techniques ensure that overlays (e.g., tooltips, highlights) do not block underlying data marks.

5. Annotations

   - Contextual annotations highlight key events (e.g., policy changes, peak months) with text and coloured marks.
   - Tooltips appear on hover for most charts, delivering high data density without overwhelming the static display.
   - For the map, selection highlights the chosen region with increased saturation and brings up a key region summary in a layered popup.

3.3 Interaction Design

Interaction in the dashboard is designed to be intuitive and support fluid exploratory analysis without clutter or occlusion. All interactions are implemented using accessible, visually distinct channels (hue, saturation, shape) and maintain high data-to-ink ratio.

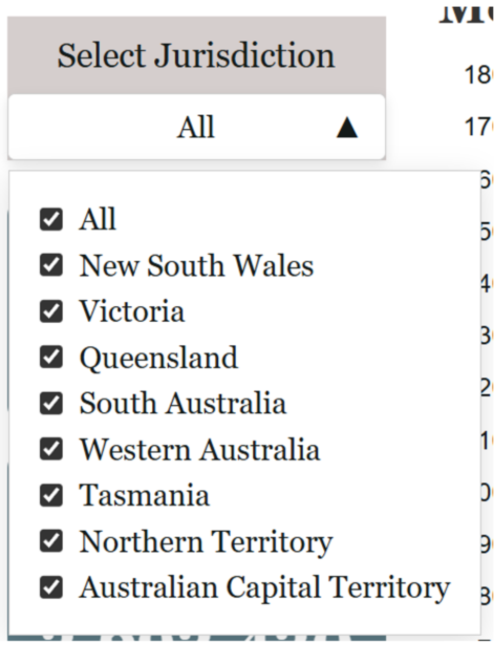

Figure 14. Filter button with its dropdown list

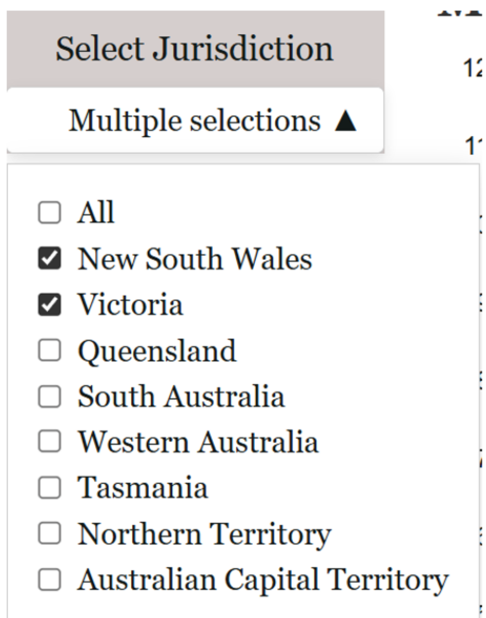

Figure 15. When filtering button only New South Wales and Victoria

Figure 16. Card charts show only data from New South Wales and Victoria

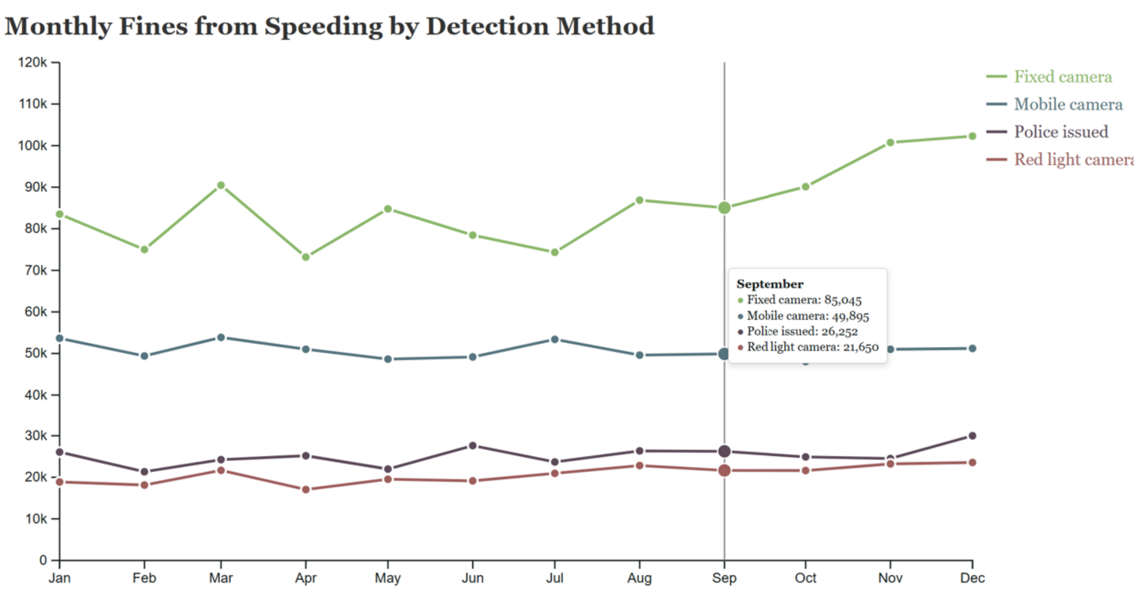

Figure 17. Line chart shows only data from New South Wales and Victoria

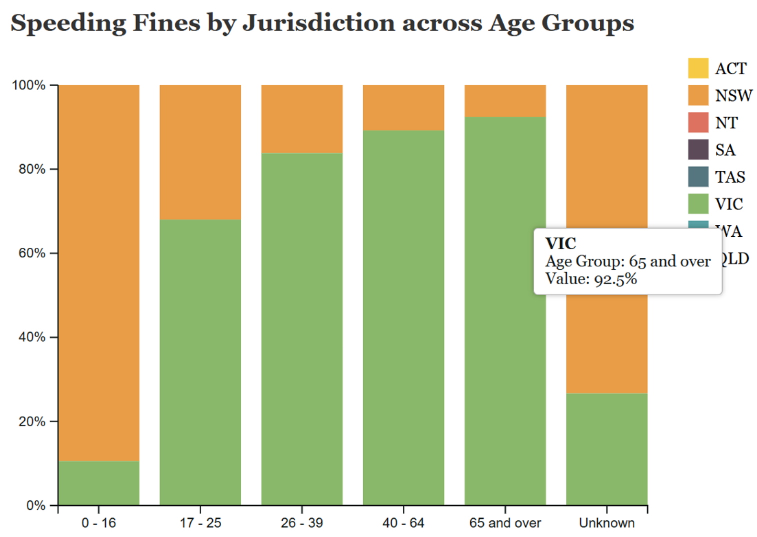

Figure 18. Stacked Bar chart shows only data from New South Wales and Victoria

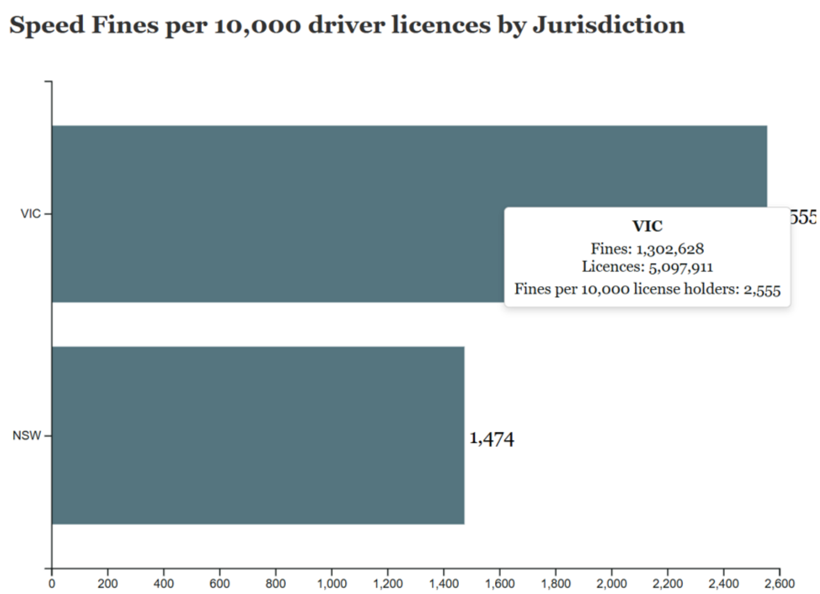

Figure 19. Horizontal Bar chart shows only data from New South Wales and Victoria

Summary of Interactive Features

Interaction Method	User Action	Expected Response
Hover	Move cursor over any chart element (tick/mark)	Tooltip appears, showing precise attribute values (e.g., fine amount, jurisdiction)
Click (Map)	Click on a jurisdiction in the choropleth map	Map zooms to selected region and overlays a summary popup with detailed statistics
Dropdown	Select or deselect jurisdiction(s) in filter	All charts except the map update dynamically, showing only data for selected jurisdictions
Button	Click "Select All" or "Unselect All" in filter	All charts reset to the full dataset or clear the selection, restoring default view

Interaction Notes

- The filter function applies to card charts, the line chart, stacked bar chart, and horizontal bar chart. It does not affect the choropleth map, ensuring spatial integrity and avoiding incomplete map regions.
- Each chart uses tooltips (activated by hover) for data exploration, maximising data density without increasing visual clutter.
- Visual feedback (such as subtle changes in hue, saturation, or border) provides clear indication of interactive elements and selections.
- The interaction design employs spatial separation between filters and charts to avoid accidental occlusion and to guide user attention.

Accessibility and Responsiveness

All interactive controls are designed to be accessible (suitable for keyboard and mouse/touch 1.

4 Iteration and Validation

4.1 Testing and refinements

This project followed an iterative process, improving the dashboard at each stage based on both technical testing and feedback from users, peers, and the teaching team. The core goal was to ensure that the visualisation was not only visually appealing and information-rich, but also accessible, intuitive, and robust across all supported devices.

Early Version Limitations and Initial Feedback

Early prototypes had several issues, identified through self-testing and peer review:

- **Line Chart:** The initial version did not have interactive tooltips or a dynamic y-axis. End-of-line labels often overlapped, especially for methods with similar values, making the chart difficult to read.

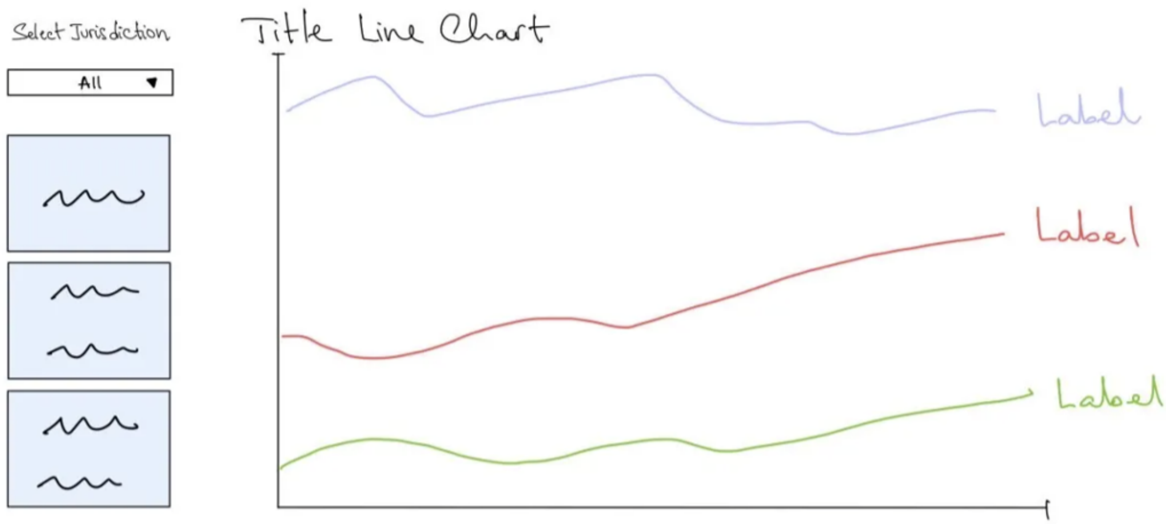

Figure 20. Early Version of the Card charts and the Line Chart

- **Stacked Bar Chart:** No tooltips were provided, and the colour palette was not suitable for users with colour vision deficiency, limiting accessibility.

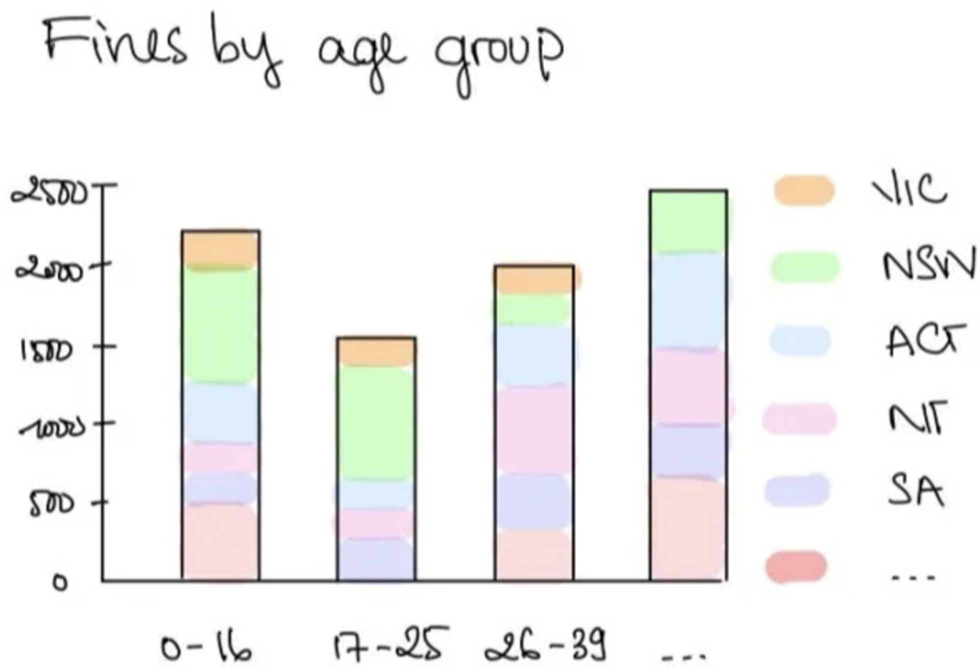

Figure 21. Early Version of the Stacked Bar Chart

- **Horizontal Bar Chart:** The early version lacked tooltips, and the bar colours did not align with the final, accessible palette.

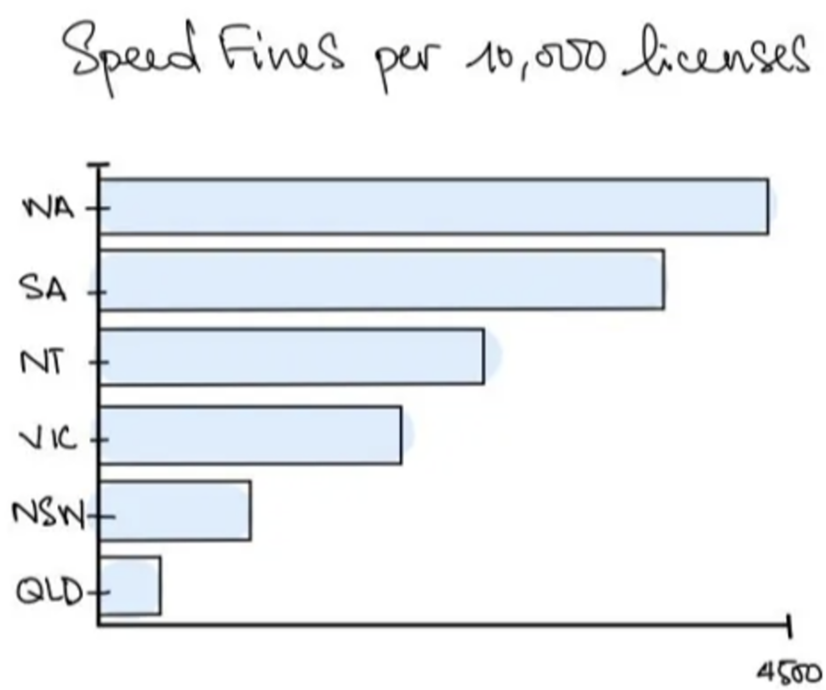

Figure 22. Early Version of the Horizontal Bar Chart

- **Choropleth Map:** The early map lacked a tooltip table, did not support zooming on jurisdictions, and the colour gradient legend was misleading (0–100%, though the maximum was only 31%). The ACT was hard to identify on the map and not labelled clearly.

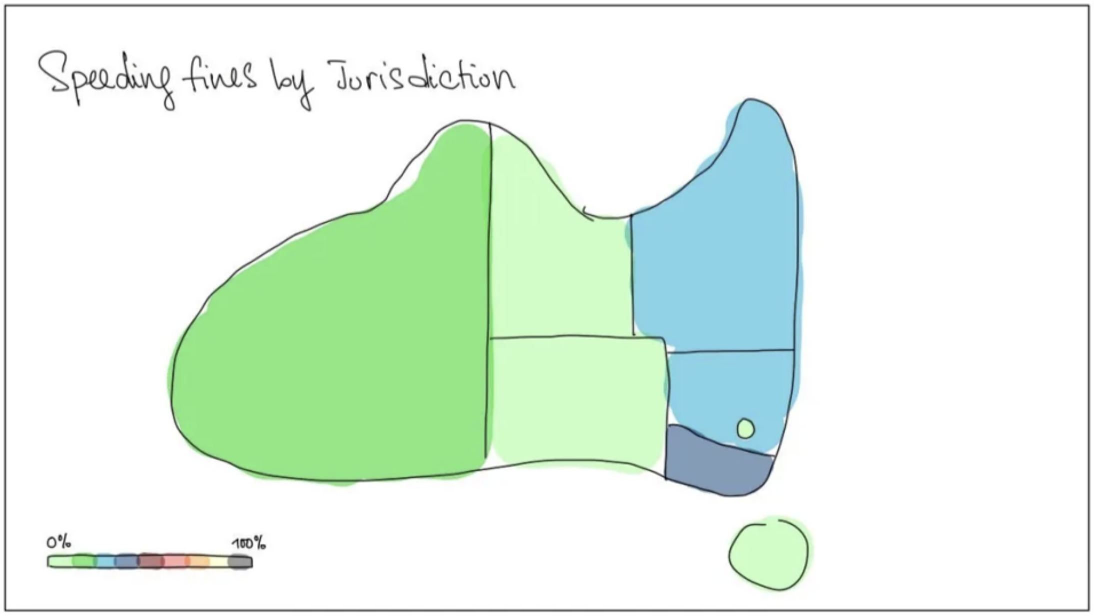

Figure 23. Early Version of the Choropleth Map Chart

These figures (20-23) illustrate the shortcomings of the initial design and are included for comparison with the final versions already referenced in Section 3.

Refinements and Final Version

Based on identified issues and user feedback, the following improvements were implemented:

- Universal Tooltips: Tooltips are now implemented on all charts, making it easy to explore details interactively.
- Line Chart Improvements:
  - Dynamic Y-Axis automatically scales with the filter selection.
  - Legend Placement is now on the right, eliminating overlapping labels.
- Stacked and Horizontal Bar Charts:
  - Updated with colourblind-friendly palettes and accessible legends.
  - Tooltips now display key details on hover.
- Choropleth Map Enhancements:
  - Tooltip table and zoom-in on jurisdiction click.

    - The colour gradient legend matches the data distribution, and ACT is clearly labelled.

- Responsiveness and Layout: CSS Flexbox and Grid ensure all components adapt across screen sizes, and interactive controls remain accessible.

- Accessibility: Colours tested for colour vision deficiency and improved keyboard navigation.

Testing Process

Testing was performed at each development milestone:

- Visualisation features were tested by filtering for single/multiple jurisdictions, ensuring all charts updated instantly and correctly.

- Peer testers and the teaching team gave feedback during demonstration sessions (e.g., line label overlaps, missing tooltips).

- Manual testing confirmed proper rendering on both desktop and mobile, with legends and controls remaining visible and accessible.

Before-and-After Comparison

Each major refinement was guided by user feedback and design best practices. Figures 16–19 (above) demonstrate the limitations of the early versions, while Figures 11–15 (already shown in Section 3) illustrate the clarity and usability of the final dashboard.

4.2 Usability evaluation

A usability evaluation was conducted with three peer reviewers and one teaching team member, all with some familiarity with data analysis or visualisation. Reviewers were asked to complete tasks relevant to the dashboard's goals.

Evaluation Tasks

Participants completed the following:

1. Identify Trends: "Use the line chart to find which detection method saw the most fines in June 2023."

2. Compare Jurisdictions: "Filter for Victoria and Queensland only. Which state had more fines per 10,000 licence holders?"

3. Age Group Analysis: "In New South Wales, which age group received the highest proportion of speeding fines?"

4. Map Interaction: "Click on a state in the map to see its share of national fines and test the zoom and tooltip."

Key Findings

- All users successfully completed the tasks.
- The new legend and tooltips were praised for clarity.
- The jurisdiction filter was described as intuitive and responsive.
- One reviewer initially missed the map's click ability, prompting an added instruction above the map.
- The colour palette—selected and tested with ColorBrewer guidelines—was found to be readable by all, including users testing with a colour-blindness simulator.
- Accessibility for keyboard navigation was confirmed, but one suggestion was made to increase the filter checkbox size on mobile.

Adjustments Based on Usability Feedback

- Increased clickable area for filter checkboxes.
- Increased clickable area for filter checkboxes.

Through iterative design and usability testing, the dashboard became significantly more functional and user-friendly. Early version shortcomings (Figures 16–19) were addressed in the final version (Figures 11–15). The use of ColorBrewer palettes and accessibility-focused testing ensured that visual distinctions are clear for all users, including those with colour vision deficiencies. Feedback from users was directly incorporated, resulting in a dashboard that is accessible, visually clear, and engaging.

5 Conclusion and Future Improvements

5.1 Conclusion

This project is the journey of applying what we have learnt throughout the semester to explore the distribution of speeding fines and enforcement actions across Australian jurisdictions. It is separated into three main phases: cleaning and exploring the data in KNIME, planning our visual approach in Power BI, and finally, bringing it all to life through interactive visualisations with D3.js.

One of the key insights from our work was just how dominant fines are compared to arrests and charges, which were both rare and heavily skewed. Along the way, we encountered some challenges – most notably with Queensland's data. Unlike other jurisdictions, Queensland doesn't report enforcement actions by age group or by month, which made it difficult to include in time-based comparisons. To work around this, we decided to evenly divide Queensland's data across months to maintain consistency.

The interactive visualisations we built—from line charts to map-based views—helped us better understand and compare enforcement patterns across regions and age groups. They also reinforced how important it is to use proportion-based views when working with data from areas with very different population sizes.

Throughout the project, we learned the importance of clean, structured data, the impact of visual design on narrative clarity, and the technical complexity of building effective, scalable visualisations in D3.js.

5.2 Future Improvements

This dashboard is currently just simply visualising past data. There are a few ways we could take this project further and make it even stronger.

One key improvement would be to integrate real-time data or regularly update datasets, allowing users to track ongoing trends rather than just relying on static 2023 data.

Since we both have experienced in machine learning, we agreed that another enhancement could involve adding predictive analytics – for example, using machine learning models to forecast future fine trends based on seasonal patterns.

Lastly, we could integrate some supplementary datasets such as weather conditions or public events, which may reveal interesting correlations with spikes in enforcement actions.

6 Appendix 1: GenAI Usage

Generative AI tools such as ChatGPT, Gemini, and Copilot were utilized with a clear understanding to enhance the development of this project. The following examples illustrate specific prompts posed and the corresponding guidance received:

1. Prompt: How can a Cartesian product be performed between two tables in KNIME?
   Response: ChatGPT identified that the Cross Joiner node within KNIME executes a Cartesian product (cross join) between two input tables. Additional explanations on configuration and usage were provided to clarify the process.

2. Prompt: How can the D3.js code be modified to animate transitions during data filtering or updates, such as enabling bars to fade and resize smoothly?
   Response: ChatGPT recommended incorporating the .transition().duration(750) method when updating chart elements, thereby facilitating smooth animation effects during state changes.

3. Prompt: What modifications can be applied to a D3 visualization to improve accessibility for users with colour vision deficiencies or those relying on screen readers?
   Response: The suggestions include:
   Employing colourblind-friendly palettes, such as ColorBrewer's "Colourblind Safe" schemes, which are integrated within the d3-scale-chromatic library, to ensure distinguishability for colour-impaired users.
   Adding textual labels or numeric values directly on the visual elements, reducing dependence on colour cues alone and enhancing interpretability for screen reader technologies.

7 Appendix 2: Resources Used

D3.js in Action, Third Edition

Australia map, by Gerardo Furtado

10 Data Visualization Best Practices for Improved Decision-Making, by Luzmo

ColorBrewer: Color Advice for Maps

8 Reference